Mostafa Ismail
--**** | Email | LinkedIn | GitHub | Google Scholar

Summary

- PhD biomedical/electrical engineer (University of Pennsylvania, Functional and Metabolic Imaging Group).
- Specializing in computer vision (AI/ML), signal/image processing, and machine/deep learning for medical imaging and image analysis (2D/3D , MRI/CT/X-ray).
- Strong background in medical imaging, computer vision, and machine perception algorithms (camera calibration, homography, bundle adjustment, 3D medical image reconstruction, FFTs, k-space, etc).
- Comfortable at the integration of hardware and software to design experiments, collect novel datasets, and iterate on prototypes.
- Skills: Python (PyTorch, NumPy, SciPy, pandas, OpenCV, nibabel, pydicom); medical image I/O (DICOM, NIfTI); C/C++, Github, Bash, Linux, Spark, MATLAB, LTSPICE, Altium, Modelsim

Education

University of Pennsylvania — Philadelphia, PA
PhD in Bioengineering | Functional and Metabolic imaging Group — 2020—present

Thesis: Advanced Imaging and Artificial Intelligence for Optimizing Lung Transplantation

- Designed signal processing and unsupervised machine learning algorithms to reduce motion artifacts in free-breathing micro-CT images.
- Processed NMR and MRI signals in python, including signal extraction, offset removal, filtering, peak fitting, clustering, reconstruction, and modeling.
- Built deep learning and computer vision frameworks for chest x-ray lung assessment.
- Developed processing pipeline for image segmentation and registration for large CT dataset.
- Analyzed CT and MRI imaging data to characterize and early detect lung disorders.

University of Pennsylvania — Philadelphia, PA
MS in Bioengineering — 2020—2024
GPA: 3.95/4.0

Coursework: Computer Vision, Machine Perception, Digital Signal Processing, Machine/ Deep Learning, Biomedical Image Analysis, Techniques of MRI, Linear Optimization , Molecular Imaging, Statistics.

Drexel University — Philadelphia, PA
BS / MS in Electrical Engineering — 2015—2020
GPA: 3.93/4.0, Summa Cum Laude

Coursework: Design with Embedded Processors, Analog and Digital Communications, Power Electronic Converters, Computing and Control, Electronic Devices, Digital Logic Design, Signals and Systems.

Experience

NeuralTrak — San Jose, CA
Algorithm Engineering Internship — Feb 2026—June 2026

- Led development and optimization of image stitching and geometric alignment algorithms for multi-scan volumetric reconstruction to improve anatomical coverage and spatial consistency.
- Developed 2D/3D imaging and tomosynthesis reconstruction algorithms, with a focus on image quality, artifact suppression, and CT-like visual fidelity.
- Built command-line interface (CLI) evaluation tools using Dice, MAE, and image-quality metrics to validate reconstructed volumes against reference CT/CBCT data.

University of Pennsylvania — Philadelphia, PA
Research Fellow | Functional and Metabolic imaging Group — 2020—present

- **An unsupervised approach for projection binning to reduce motion artifacts in free-breathing animal models. (**publication**)**

- Modified K-means clustering with periodic boundary conditions and utilized it for projections binning, achieving significant reduction in motion and streak artifacts.
- Extracted, filtered, and processed breathing signals from micro-CT images.

• **AI-driven automated lung sizing from chest radiographs (**PyPI package**,** publication**)**
- Created a framework for automatic measurement of lung sizes by 1) training state-of-the-art segmentations models to segment lung from chest x-ray images, and 2) Implementing computer vision techniques to identify feature points on the contour of the segmented lungs.

• **A deep learning approach for chest radiograph analysis of primary graft dysfunction after lung transplantation (**publication**)**
- Segmented left/right lungs with SOTA models, then used representation learning (CNN latent embeddings and PCA) to detect/quantify infiltrates and improve standardized PGD grading and prognosis.

• **CT-Lung-Analysis: Automated DICOM-to-lung-segmentation pipeline (**GitHub project**)**
- Built an end-to-end Python tool that converts raw DICOM chest CT series to NIfTI, segments lungs with a non-deep learning technique, and computes air-content maps for rapid quantitative assessment.
- Packaged the workflow as a reproducible notebook and executable file, integrating efficient pre-/post-processing (SimpleITK, nibabel, matplotlib) and providing robust command-line and GUI-free deployment.

• **Adaptive Filters and Equalizers for Signal Denoising and Channel Compensation**
- Implemented real-time interference cancellation and multi-interference rejection in MATLAB using adaptive IIR notch filters (LMS) and blind/constant-modulus equalization to achieve full digital signal recovery.

KavoKerr, Danaher Hatfield, PA
Electrical Engineering Intern 2017—2018

• **Class II Dental Imaging (X-RAY) device life test**
- Used Altium Designer in Schematic capture and PCB layout for designing the board. The board is a cycler board and a data logger that is used for the lifetime test for the product.
- Built and debugged the PCB.
- Programmed microcontroller to control the cycler/data logger test fixture, record data to SD card, and display data on LCD.

• **Embedded System for Smart Battery Interface Board**
- Programmed the functional tester to check the functionality of the BIB (Battery Interface Board) that is used on x-ray portable devices. The test includes various communication protocols.

• **X-ray Tubehead Radiation Leakage Scanner**
- Designed a system that scans the radiation leakage of X-ray tubes
- The design included stepper motors, sensors, embedded system, and GUI application.
- The system is used in the Research and development department for developing and testing new Imaging Device.

Leadership Experience

University of Pennsylvania Philadelphia, PA
Resident Advisor | College Houses and Administrative Services 2021—2025

- Organized and led biweekly community-building events to foster inclusive, engaging living environments for over 30+ residents.
- Provided leadership in crisis management, conflict resolution, and policy enforcement while maintaining a supportive residential community.

Drexel University Philadelphia, PA
Academic Tutor | College Houses and Administrative Services 2017—2020

- Tutored Signals and Systems, Electric Circuits, Digital Logic Design, Linear and Dynamic Engineering Systems, and Programming.
- Ran the lab sessions for the course: Introductory Programming for Engineers.

Publications

Ismail, Mostafa K., Kai Ruppert, Marco Caballo, Hooman Hamedani, Ian Duncan, Stephen Kadlecek, and Rahim Rizi. "An unsupervised approach for projection binning to reduce motion artifacts in free-breathing animal models." Medical Physics 52, no. 6 (2025): 4318-4329.

Ismail, Mostafa K., Tetsuro Araki, Warren B. Gefter, Yoshikazu Suzuki, Allie Raevsky, Aya Saleh, Sophia Yusuf et al. "Artificial intelligence-driven automated lung sizing from chest radiographs." American Journal of Transplantation 25, no. 1 (2025): 198-203.

Ruppert, Kai, Luis Loza, Hooman Hamedani, **Mostafa Ismail**, Jiawei Chen, Ian F. Duncan, Harrilla Profka, Stephen Kadlecek, and Rahim R. Rizi. "Regional variations in hyperpolarized 129Xe lung MRI: Insights from CSI-CSSR and CSSR in healthy and irradiated rat models." Magnetic Resonance in Medicine 93, no. 3 (2025): 902-915.

Selected Poster Presentations

Ismail, M. K., et al. "Assessing Post-Transplant Lung Function Alterations Using Free-Breathing Hyperpolarized Xenon MRI." American Journal of Respiratory and Critical Care Medicine 211.Abstracts (2025): A7932-A7932.

Ismail, M. K., et al. "Automated Detection of Edema Post Lung Transplant From Chest X-rays." American Journal of Respiratory and Critical Care Medicine 211.Abstracts (2025): A4740-A4740.

Ismail, M., et al. "Temporal and Volumetric Analysis of Lung Ventilation After Transplantation in a Rat Model Using Dynamic Micro-CT Imaging." The Journal of Heart and Lung Transplantation 44.4 (2025): S732-S733.

Ismail, M. K., et al. "Deep Learning for Grading Primary Graft Dysfunction from Chest Radiographs." The Journal of Heart and Lung Transplantation 44.4 (2025): S512.

Ismail, M. K., et al. "Increased Ventilation Heterogeneity in Pre-clinical Thoracic Insufficiency Syndrome Model." B77. NEW APPROACHES IN LUNG VENTILATION. American Thoracic Society, 2024. A4448-A4448.

Ismail, M. K., et al. "Optimizing Spatiotemporal Resolution in Dynamic Imaging Using Phase Binning-guided K-means Clustering." C29. ADVANCED LUNG IMAGING; THE PULMONARY PAPARAZZI. American Thoracic Society, 2024. A5208-A5208.

Ismail, M. K., et al. "Automated Measurements of Lung Sizes Using Deep Learning and Computer Vision." C23. MACHINE LEARNED: AI-DRIVEN SOLUTIONS IN ILD AND LUNG TRANSPLANT. American Thoracic Society, 2024. A5098-A5098.

Ismail, M. K., et al. "Evaluation of BM4D on Denoising Gas and Dissolved Phases of Hyperpolarized-Xenon MRI." B80-1. METHODOLOGICAL ADVANCEMENTS IN PULMONARY IMAGING. American Thoracic Society, 2024. A4497-A4497.

Ismail, M. K., et al. "Improved Dynamic Micro-CT Imaging Binning for Optimized Image Reconstruction." A69. AN IMAGE'S WORTH: STUDIES IN LUNG IMAGING. American Thoracic Society, 2023. A2292-A2292.

Ismail, M. K., et al. "Non-invasive Assessment of Post-transplant Alterations in Pulmonary Ventilation Using Dynamic 4DCT." A69. AN IMAGE'S WORTH: STUDIES IN LUNG IMAGING. American Thoracic Society, 2023. A2291-A2291.

Ismail, M. K., et al. "Dynamic 129Xe-Hyperpolarized MRI for Non-invasive Evaluation of Regional Lung Ventilation in Thoracic Insufficiency Syndrome." A69. AN IMAGE'S WORTH: STUDIES IN LUNG IMAGING. American Thoracic Society, 2023. A2290-A2290.

Honors and Awards

Lung Transplant Foundation ATS Award	2025
American Thoracic Society Abstract Award	2024 / 2025
Educational Stipend, ISMRM	2023 / 2024
Spot Award, KaVo Kerr	2017 / 2018
Drexel Scholarship	2015—2020
Third place, National Intel Science and Engineering Fair	2015
Physics Poster Award, Asian Science Camp, Singapore	2014